

MH3100 Real Analysis I

Sample Midterm Test (8 October 2025) (Solutions)

Official sample paper – QRS solutions (Practice 1)

Quantitative Research Society @NTU

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Overview

- **Coverage:** Lectures 1–17 (real numbers and the Axiom of Completeness, through sequences, series, and sets of real numbers – limit points, closures, closed sets, and (sequential) compactness).
- **Format: FOUR (4)** questions, 6 marks each, 24 marks total. **TIME ALLOWED: 60 minutes. CLOSED BOOK.** Calculators are allowed.
- **Citation policy (verbatim from the paper):** “You may cite any results proved in the lectures and tutorials, unless you are asked by the question to prove the result.”
- **This document:** full QRS solutions to the (SAMPLE) Midterm Test dated 8 October 2025, published as *Practice 1 (Solution by QRS)*. Every proof below is written to the course’s own marking expectations (complete sentences, logical connectives, and explicit inference words), even where the printed paper itself does not require justification (Questions 1–2).

Instructions to Students (verbatim from the cover page)

1. This test contains **FOUR (4)** questions.
2. Answer **all** questions. The marks for each question are indicated below.
3. Write your answers in the box provided. Use the **facing page** for more space.
4. This is a **CLOSED BOOK** test.
5. You may use calculators. You may cite any results proved in the lectures and tutorials, unless you are asked by the question to prove the result.

Question	1	2	3	4	Total
Total Marks	6	6	6	6	24

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Question 1**(6 Marks)**

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks, QRS assumption) There are sequences $\{a_n\}$ and $\{b_n\}$, where $\{a_n\}$ converges, $\{b_n\}$ diverges, and $\{a_n + b_n\}$ converges.
- (b) (2 marks, QRS assumption) There is a sequence with a subsequence that is bounded but contains no subsequence that converges.
- (c) (2 marks, QRS assumption) If every proper subsequence of a sequence $\{a_n\}$ converges, then $\{a_n\}$ converges as well.

Solution

- (a) False (Source: Tutorial 2, Question 1(e).)

Justification. Suppose, for contradiction, that such sequences $\{a_n\}$ and $\{b_n\}$ exist, with $\{a_n\}$ convergent, $\{a_n + b_n\}$ convergent, and $\{b_n\}$ divergent. Since $\{a_n\}$ converges, the Algebraic Limit Theorem (part 1, with $c = -1$) gives that $\{-a_n\}$ converges. Since $\{a_n + b_n\}$ and $\{-a_n\}$ both converge, the Algebraic Limit Theorem (part 2) gives that their sum

$$\{(a_n + b_n) + (-a_n)\} = \{b_n\}$$

converges. This contradicts the assumption that $\{b_n\}$ diverges. Therefore no such pair of sequences exists, and the statement is false. □

- (b) False (Source: Tutorial 3, Question 2(b).)

Justification. Suppose, for contradiction, that a sequence $\{a_n\}$ has a subsequence $\{a_{n_k}\}$ that is bounded but has no convergent subsequence. Since $\{a_{n_k}\}$ is a bounded sequence, the Bolzano-Weierstrass Theorem asserts that it has a convergent subsequence $\{a_{n_{k_j}}\}$. A subsequence of a subsequence of $\{a_n\}$ is itself a subsequence of $\{a_n\}$, since the composition of two strictly increasing sequences of natural numbers is again strictly increasing; in particular, $\{a_{n_{k_j}}\}$ is a subsequence of $\{a_{n_k}\}$ that converges. This contradicts the assumption that $\{a_{n_k}\}$ has no convergent subsequence. Therefore every sequence with a bounded subsequence has, within that bounded subsequence, a further convergent subsequence, and the statement is false. □

- (c) True (Source: Tutorial 3, Question 1(a).)

Idea. (Scratch work – not part of the written answer.) By the course's Definition (Subsequence), a subsequence $\{a_{n_k}\}$ of $\{a_n\}$ is *proper* if $n_k \neq k$ for some $k \in \mathbb{N}$. In particular, the “identity” subsequence $n_k = k$ for all k (i.e. $\{a_n\}$ itself) is *not* proper, so the hypothesis says nothing directly about $\{a_n\}$ as a whole. However, deleting a single term from $\{a_n\}$ already produces a proper subsequence: for any fixed index $m \in \mathbb{N}$, the sequence obtained by omitting a_m has index sequence $n_k = k$ for $k < m$ and $n_k = k + 1$ for $k \geq m$, which differs from the identity at every $k \geq m$, so it is proper. The key trick is to delete two *different* single terms and compare the (identical) tails of the two resulting proper subsequences.

Justification. Suppose every proper subsequence of $\{a_n\}$ converges. Let $\{b_k\}$ denote the sequence obtained by deleting a_1 , i.e. $b_k := a_{k+1}$ for each $k \in \mathbb{N}$; its index sequence is $n_k = k + 1 \neq k$ for every k , so $\{b_k\}$ is a proper subsequence of $\{a_n\}$, and by hypothesis it converges to some limit $L \in \mathbb{R}$. Let $\{c_k\}$ denote the sequence obtained by deleting a_2 instead, i.e. $c_1 := a_1$ and $c_k := a_{k+1}$ for $k \geq 2$; its index sequence satisfies $n_2 = 3 \neq 2$, so $\{c_k\}$ is also a proper subsequence of $\{a_n\}$, and by hypothesis it converges to some limit $M \in \mathbb{R}$.

Consider the sequence $a_3, a_4, a_5, \dots$. This is simultaneously a subsequence of $\{b_k\}$ (namely $b_2, b_3, b_4, \dots$) and a subsequence of $\{c_k\}$ (namely $c_2, c_3, c_4, \dots$). By the theorem on Subsequential Limits, every subsequence of a convergent sequence converges to the same limit as the original sequence; hence $a_3, a_4, a_5, \dots$ converges to L (as a subsequence of $\{b_k\}$) and also to M (as a subsequence of $\{c_k\}$). Since limits of sequences are unique, $L = M$.

It remains to show that $\{a_n\}$ itself converges to L . Since $\{b_k\} = a_2, a_3, a_4, \dots$ converges to L , the definition of $L = \lim b_k$ gives: for every $\varepsilon > 0$ there exists $K \in \mathbb{N}$ such that for all $k \geq K$, $|a_{k+1} - L| < \varepsilon$. Setting $N := K + 1$, this says that for all $n \geq N$ (writing $n = k + 1$), $|a_n - L| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, this is exactly the definition of $L = \lim a_n$. Therefore $\{a_n\}$ converges (to L), and the statement is true. $\square$

Remark. The argument used only two of the (infinitely many) proper subsequences of $\{a_n\}$ – namely, the two obtained by deleting a single early term – together with the theorem on Subsequential Limits and the basic fact that convergence of a sequence is unaffected by discarding finitely many leading terms. No special property of “1” and “2” was used, so the same argument works with any two distinct deleted indices.

Common slip. Treating $\{a_n\}$ itself as one of the “proper” subsequences covered by the hypothesis in part (c) (it is not, since its index sequence is the identity $n_k = k$); or, conversely, assuming without proof that the statement must be false simply because the hypothesis excludes the sequence itself.

Marking scheme (6 marks)

QRS suggested marking scheme (unofficial): the printed paper awards these 6 marks purely for the three correct True/False answers, since “You do NOT need to justify your answer.” The scheme below is what QRS would use if justification were required, for teaching purposes.

(a) **(2 marks)** Correct answer False.

- 2 marks: correct answer. (If justification were required: 1 mark for the answer, 1 mark for correctly invoking the Algebraic Limit Theorem to derive a contradiction.)

(b) **(2 marks)** Correct answer False.

- 2 marks: correct answer. (If justification were required: 1 mark for the answer, 1 mark for correctly invoking the Bolzano-Weierstrass Theorem.)

(c) **(2 marks)** Correct answer True.

- 2 marks: correct answer. (If justification were required: 1 mark for correctly identifying that deleting a single term produces a proper subsequence, 1 mark for the shared-tail argument forcing $\{a_n\}$ itself to converge.)

Typical deductions if justification were graded: -1 : claims $\{a_n\}$ is itself one of the “proper” subsequences in part (c); -1 : asserts the Bolzano-Weierstrass Theorem without first noting

boundedness of the subsequence in part (b); -0.5 : sentence fragments in place of a full justification.

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Question 2

(6 Marks)

Is each of the following statement true or false? You do NOT need to justify your answer.

- (a) (2 marks, QRS assumption) There are two divergent series $\sum a_n$ and $\sum b_n$ where $\sum a_n b_n$ converges.
- (b) (2 marks, QRS assumption) The set $\{x \in \mathbb{R} : x \neq 0\}$ is closed.
- (c) (2 marks, QRS assumption) If K is a compact set and F is a closed set, then $K \setminus F$ is a compact set.

Solution

- (a) True (Source: Tutorial 4, Question 2(a).)

Justification. Take $a_n := 1/n$ and $b_n := 1/n$ for each $n \in \mathbb{N}$. The series $\sum a_n = \sum 1/n$ is the harmonic series, which diverges (a standard, citable fact from Lecture 10, used throughout Tutorials 4–5; e.g. grouping the terms $\frac{1}{2^{k+1}} + \cdots + \frac{1}{2^{k+1}} \geq \frac{1}{2}$ for each $k \in \mathbb{N}$ shows the partial sums are unbounded). Hence $\sum b_n = \sum 1/n$ diverges as well. However,

$$\sum a_n b_n = \sum \frac{1}{n^2}$$

converges: for $n \geq 2$, $\frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$, and the series $\sum_{n \geq 2} \left(\frac{1}{n-1} - \frac{1}{n}\right)$ has partial sums $1 - \frac{1}{n} \rightarrow 1$, so it converges; the Comparison Test (applied to $n \geq 2$, together with the single extra term at $n = 1$) then gives that $\sum 1/n^2$ converges. Thus $\sum a_n$ and $\sum b_n$ are two divergent series with $\sum a_n b_n$ convergent, and the statement is true. $\square$

- (b) False (Source: repackages Tutorial 5, Question 1(c).)

Justification. Let $C := \{x \in \mathbb{R} : x \neq 0\}$. Consider the sequence $a_n := 1/n$ for $n \in \mathbb{N}$. Since $1/n \neq 0$ for every $n \in \mathbb{N}$, the sequence $\{a_n\}$ is contained in C ; and $\{a_n\}$ converges to 0. However, $0 \notin C$. Thus $\{a_n\}$ is a convergent sequence contained in C whose limit does not lie in C , so by the Sequential Criterion for Closed Sets, C is not closed. Therefore the statement is false. $\square$

- (c) False (Source: new combination, L13–17.)

Idea. (Scratch work – not part of the written answer.) A single counterexample suffices. Take $K = [0, 1]$ (closed and bounded, hence compact) and $F = \{0\}$ (a single point, hence closed). Then $K \setminus F = (0, 1]$, and $(0, 1]$ is not compact because the sequence $1/n$ inside it converges only to 0, which has been removed.

Justification. Let $K := [0, 1]$ and $F := \{0\}$. The set F is finite, hence has no limit points at all, so it vacuously contains all of its limit points; therefore F is closed. The set K is closed and bounded, so by the Theorem (Sequential Compactness of Closed and Bounded Sets), K is (sequentially) compact.

Now $K \setminus F = (0, 1]$. Consider the sequence $a_n := 1/n$ for $n \in \mathbb{N}$; every term satisfies $0 < 1/n \leq 1$, so $\{a_n\}$ is contained in $(0, 1]$. The sequence $\{a_n\}$ converges to 0, and by the theorem on Subsequential Limits, every subsequence of $\{a_n\}$ converges to the same

limit 0. Since $0 \notin (0, 1]$, no subsequence of $\{a_n\}$ converges to a limit in $(0, 1]$. By the Definition ((Sequentially) Compact Set), this shows that $(0, 1] = K \setminus F$ is not compact.

Hence K is compact, F is closed, and $K \setminus F$ is not compact, so the statement is false. $\square$

Common slip. Confusing “closed” with “compact” in part (c) and reasoning only about closedness of $K \setminus F$ (which is in fact also false in general, since the difference of a closed set and an open complement need not be closed unless boundedness is separately tracked) instead of exhibiting an explicit witnessing sequence; or asserting $K \setminus F$ is compact “because K is compact and removing points cannot add limit points,” which ignores that removing a limit point of $K \setminus F$ from K destroys closedness of the difference.

Marking scheme (6 marks)

QRS suggested marking scheme (unofficial): as in Question 1, the printed paper awards these 6 marks purely for the three correct True/False answers. The scheme below is what QRS would use if justification were required.

(a) **(2 marks)** Correct answer .

- 2 marks: correct answer. (If justification were required: 1 mark for a valid example with both $\sum a_n, \sum b_n$ divergent, 1 mark for correctly showing $\sum a_n b_n$ convergent, e.g. via the Comparison Test.)

(b) **(2 marks)** Correct answer .

- 2 marks: correct answer. (If justification were required: 1 mark for exhibiting a convergent sequence in C with limit outside C , 1 mark for correctly invoking the Sequential Criterion for Closed Sets.)

(c) **(2 marks)** Correct answer .

- 2 marks: correct answer. (If justification were required: 1 mark for a valid counterexample (K, F) with K compact and F closed, 1 mark for correctly showing $K \setminus F$ fails compactness via an explicit non-convergent-in-the-set sequence.)

Typical deductions if justification were graded: -1 : in part (a), an example where $\sum a_n b_n$ does not actually converge (e.g. forgetting to check the product series, not just the individual terms); -1 : in part (c), a counterexample where F is not actually closed or K is not actually compact; -0.5 : sentence fragments in place of a full justification.

Question 3

(6 Marks)

In this question, A and B denote nonempty sets of real numbers that are bounded above.

Write down the definition of $s = \sup A$, and use the definition to show that

$$\max\{\sup A, \sup B\} = \sup(A \cup B).$$

Do not use the ε -Criterion for suprema and infima.

(Source: Tutorial 1, Question 2, with the added instructions to first state the definition of supremum and to avoid the ε -Criterion.)

Solution

Idea. (Scratch work – not part of the written answer.) The claim is an “if and only if” pair of inequalities in disguise: $\max\{\sup A, \sup B\}$ should be shown to be an upper bound of $A \cup B$ (giving $\sup(A \cup B) \leq \max\{\sup A, \sup B\}$, via the leastness of $\sup(A \cup B)$), and every upper bound of $A \cup B$ should be shown to bound $\max\{\sup A, \sup B\}$ from above (giving $\max\{\sup A, \sup B\} \leq \sup(A \cup B)$). Working directly from the two-part definition of supremum, rather than the ε -Criterion, packages both directions into a single “ M is the supremum of $A \cup B$ ” verification.

Definition. A number $s \in \mathbb{R}$ is said to be the *supremum* of a set $A \subseteq \mathbb{R}$, written $s = \sup A$, if

- (1) s is an upper bound for A , i.e. for every $a \in A$, $a \leq s$; and
- (2) s is not greater than any upper bound of A , i.e. for every upper bound b of A , $s \leq b$.

Method 1: Direct verification of both parts of the definition

Proof. Let $s := \sup A$ and $t := \sup B$; both exist by the Axiom of Completeness, since A and B are nonempty and bounded above. Let $M := \max\{s, t\}$. We show that $M = \sup(A \cup B)$ by verifying the two parts of the definition of supremum for M and the set $A \cup B$.

First, note that $A \cup B$ is nonempty, since A is nonempty, and $A \cup B$ is bounded above by M (this is verified as part of the next paragraph). Hence, by the Axiom of Completeness, $\sup(A \cup B)$ exists.

M is an upper bound for $A \cup B$. Let $x \in A \cup B$ be arbitrary. If $x \in A$, then since $s = \sup A$ is an upper bound for A , $x \leq s$; since $s \leq M$, it follows that $x \leq M$. If instead $x \in B$, then since $t = \sup B$ is an upper bound for B , $x \leq t$; since $t \leq M$, it follows that $x \leq M$. In either case $x \leq M$. Since $x \in A \cup B$ was arbitrary, M is an upper bound for $A \cup B$.

M is not greater than any upper bound of $A \cup B$. Let $b \in \mathbb{R}$ be an arbitrary upper bound for $A \cup B$. Since $A \subseteq A \cup B$, every $a \in A$ satisfies $a \leq b$; thus b is also an upper bound for A . By part (2) of the definition of $s = \sup A$, $s \leq b$. Symmetrically, since $B \subseteq A \cup B$, b is an upper bound for B , so by part (2) of the definition of $t = \sup B$, $t \leq b$. Therefore both $s \leq b$ and $t \leq b$, so $M = \max\{s, t\} \leq b$. Since b was an arbitrary upper bound for $A \cup B$, M is not greater than any upper bound of $A \cup B$.

Both parts of the definition of supremum hold for M and $A \cup B$, so $M = \sup(A \cup B)$. That is, $\max\{\sup A, \sup B\} = \sup(A \cup B)$. $\square$

Method 2: WLOG reduction

Idea. (Scratch work – not part of the written answer.) By symmetry between A and B , we may assume without loss of generality that $\sup A \geq \sup B$, which collapses $\max\{\sup A, \sup B\}$

to the single value $\sup A$ and removes the need to case-split on which set a point $x \in A \cup B$ belongs to.

Proof. Let $s := \sup A$ and $t := \sup B$, which exist by the Axiom of Completeness. Without loss of generality, assume $s \geq t$ (the case $t \geq s$ is symmetric, with the roles of A and B exchanged); then $\max\{s, t\} = s$. We show $s = \sup(A \cup B)$.

Since A is nonempty, $A \cup B$ is nonempty; and $A \cup B$ is bounded above by s (verified below), so $\sup(A \cup B)$ exists by the Axiom of Completeness.

s is an upper bound for $A \cup B$. Let $x \in A \cup B$. If $x \in A$, then $x \leq s$ since $s = \sup A$ is an upper bound for A . If $x \in B$, then $x \leq t$ since $t = \sup B$ is an upper bound for B ; and since $t \leq s$, $x \leq s$. Hence $x \leq s$ for every $x \in A \cup B$.

s is not greater than any upper bound of $A \cup B$. Let b be an arbitrary upper bound for $A \cup B$. Since $A \subseteq A \cup B$, b is an upper bound for A ; since $s = \sup A$ is not greater than any upper bound for A , $s \leq b$.

Both parts of the definition hold for s and $A \cup B$, so $s = \sup(A \cup B)$. Since $s = \max\{s, t\} = \max\{\sup A, \sup B\}$ by the standing assumption $s \geq t$, this gives $\max\{\sup A, \sup B\} = \sup(A \cup B)$. $\square$

Common slip. Invoking $\sup(A \cup B)$ before checking that $A \cup B$ is nonempty and bounded above (the Axiom of Completeness does not apply otherwise); proving only that M is an upper bound for $A \cup B$ and stopping there, without the second, “least upper bound” direction (or vice versa); starting the second direction from “ $M \leq b$ ” as if it were already known, rather than fixing an arbitrary upper bound b of $A \cup B$ and deriving $s \leq b$ and $t \leq b$ separately from the definition of s and t ; and using the ε -Criterion for suprema and infima, which the question explicitly forbids.

Marking scheme (6 marks)

QRS suggested marking scheme (unofficial).

- (i) **(1 mark) Definition.** The definition of $s = \sup A$ stated correctly, with both the “upper bound” and “least upper bound” clauses, and quantifiers in the correct order (“for all $a \in A$, $a \leq s$ ”; “for every upper bound b of A , $s \leq b$ ”).
 - -0.5 : only one of the two clauses given.
 - -0.5 : quantifier order reversed or omitted (e.g. writing “ $s \leq b$ for some upper bound b ”).
- (ii) **(1 mark) Existence.** Noting that $A \cup B$ is nonempty and bounded above, so that $\sup(A \cup B)$ exists by the Axiom of Completeness, before using it.
 - -1 : uses $\sup(A \cup B)$ without ever justifying its existence.
- (iii) **(2 marks) Upper bound direction.** A correct proof that $\max\{\sup A, \sup B\}$ is an upper bound for $A \cup B$, with the case split $x \in A / x \in B$ (or the WLOG reduction of Method 2) fully carried out.
 - -1 : case split incomplete (only one of A , B checked).
- (iv) **(1.5 marks) Least-upper-bound direction.** A correct proof that $\max\{\sup A, \sup B\}$ is not greater than any upper bound of $A \cup B$, via an arbitrary upper bound b of $A \cup B$, restricting b to A and to B , and applying the definition of $\sup A$ and $\sup B$ to conclude $\sup A \leq b$ and $\sup B \leq b$.

- -1 : asserts $\sup A \leq \sup(A \cup B)$ (or similar) without the arbitrary-upper-bound argument.

(v) **(0.5 marks) Writing.** Full sentences, connectives, and inference words (“therefore”, “hence”, “so”) throughout, rather than a bare symbolic chain.

Typical further deductions: -1 : uses the ε -Criterion for suprema and infima, which is explicitly banned by the question; -0.5 : sentence fragments or unjustified jumps between lines.

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Question 4

(6 Marks)

Write down the definition of a Cauchy sequence $\{a_n\}$, and **use the definition to show** that if $\{a_n\}$ is a Cauchy sequence [sic], then $\{a_n^2\}$ is a Cauchy sequence.

Do not use the Cauchy Criterion or the Algebraic Limit Theorem.

(Source: Tutorial 2, Question 5(b), with the added instruction to first state the definition of a Cauchy sequence.)

Solution

Idea. (*Scratch work – not part of the written answer.*) Write $a_n^2 - a_m^2 = (a_n - a_m)(a_n + a_m)$. If $\{a_n\}$ is bounded by some $M > 0$, then $|a_n + a_m| \leq 2M$, so $|a_n^2 - a_m^2| \leq 2M |a_n - a_m|$; making $|a_n - a_m|$ small (via the Cauchy hypothesis, with target $\varepsilon/(2M)$) makes $|a_n^2 - a_m^2|$ small. The only extra ingredient is that a Cauchy sequence is bounded – and since the Cauchy Criterion (which would go via convergence) is banned, this boundedness must be proved directly from the Cauchy definition, exactly as Boundedness of Convergent Sequences is proved from the convergence definition, using $\varepsilon = 1$.

Definition. A sequence $\{a_n\}$ is said to be *Cauchy* if

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n, m \in \mathbb{N} \quad n, m \geq N \implies |a_n - a_m| < \varepsilon.$$

Method 1: Boundedness, then a factoring estimate

Lemma (a Cauchy sequence is bounded). If $\{a_n\}$ is Cauchy, then $\{a_n\}$ is bounded, i.e. there exists $M > 0$ such that $|a_n| \leq M$ for all $n \in \mathbb{N}$.

Proof of Lemma. Let $\{a_n\}$ be Cauchy. Applying the definition of Cauchy with $\varepsilon = 1$, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, $|a_n - a_m| < 1$. In particular, taking $m = N$, for all $n \geq N$, $|a_n - a_N| < 1$; by the triangle inequality,

$$|a_n| \leq |a_n - a_N| + |a_N| < 1 + |a_N|.$$

So every term with index $n \geq N$ is bounded by $1 + |a_N|$. The remaining, finitely many terms $a_1, a_2, \dots, a_{N-1}$ are each a single fixed real number, hence individually bounded. Let

$$M := \max\{|a_1|, |a_2|, \dots, |a_{N-1}|, 1 + |a_N|\}$$

(a maximum of finitely many, and at least one, real numbers, so it exists). Since $1 + |a_N| \geq 1 > 0$ is one of the numbers in this list, $M \geq 1 > 0$. By construction, $|a_n| \leq M$ for every $n \in \mathbb{N}$: this holds for $n < N$ directly from the list, and for $n \geq N$ since $|a_n| < 1 + |a_N| \leq M$. Therefore $\{a_n\}$ is bounded. $\square$

Proof of Q4. Let $\{a_n\}$ be a Cauchy sequence. By the Lemma above, $\{a_n\}$ is bounded: there exists $M > 0$ such that $|a_n| \leq M$ for every $n \in \mathbb{N}$.

Let $\varepsilon > 0$ be given. Since $M > 0$, the number $\varepsilon' := \varepsilon/(2M)$ is a well-defined positive real number. Since $\{a_n\}$ is Cauchy, applying the definition of Cauchy with ε' in place of ε , there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, $|a_n - a_m| < \varepsilon'$.

Fix $n, m \geq N$. Then, using the factoring identity $a_n^2 - a_m^2 = (a_n - a_m)(a_n + a_m)$, the triangle inequality, and the bound $|a_n|, |a_m| \leq M$,

$$|a_n^2 - a_m^2| = |a_n - a_m| \cdot |a_n + a_m|$$

$$\begin{aligned}
&\leq |a_n - a_m| (|a_n| + |a_m|) \\
&\leq |a_n - a_m| \cdot 2M \\
&< \varepsilon' \cdot 2M = \frac{\varepsilon}{2M} \cdot 2M = \varepsilon.
\end{aligned}$$

Since $n, m \geq N$ were arbitrary, this shows that for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, $|a_n^2 - a_m^2| < \varepsilon$. This is exactly the definition of $\{a_n^2\}$ being Cauchy. Therefore $\{a_n^2\}$ is a Cauchy sequence. $\square$

Common slip. Asserting that $\{a_n\}$ is bounded by quoting “Boundedness of Convergent Sequences,” which requires $\{a_n\}$ to already be known convergent – but that is precisely what the (banned) Cauchy Criterion would supply, so boundedness of a merely Cauchy sequence must be re-derived directly from the Cauchy definition with $\varepsilon = 1$, as in the Lemma; choosing N so that $|a_n - a_m| < \varepsilon$ directly (forgetting the factor of $2M$ needed to control $|a_n^2 - a_m^2|$, not just $|a_n - a_m|$); and quietly invoking the Algebraic Limit Theorem (e.g. “ $a_n^2 \rightarrow a^2$ since $a_n \rightarrow a$ ”), which both begs the question (a Cauchy sequence’s convergence has not been assumed or shown) and is explicitly banned by the question.

Marking scheme (6 marks)

QRS suggested marking scheme (unofficial).

- (i) **(1 mark) Definition.** The definition of a Cauchy sequence stated correctly, with quantifiers in the correct order: for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$, $|a_n - a_m| < \varepsilon$.
 - -0.5 : quantifier order wrong (e.g. N chosen before ε , or N allowed to depend on n).
 - -0.5 : uses the single-index form ($|a_n - a_N| < \varepsilon$) without adapting the rest of the proof to match, or silently swaps between the two forms.
- (ii) **(1.5 marks) Boundedness Lemma.** A correct, self-contained proof that a Cauchy sequence is bounded, directly from the Cauchy definition with $\varepsilon = 1$, splitting into the tail (bounded by $1 + |a_N|$) and the finitely many head terms.
 - -1 : boundedness asserted without proof, or proved via convergence/the Cauchy Criterion.
 - -0.5 : omits the finitely-many-head-terms case, or does not verify $M > 0$.
- (iii) **(2 marks) Factoring estimate.** The identity $a_n^2 - a_m^2 = (a_n - a_m)(a_n + a_m)$, the triangle-inequality bound $|a_n + a_m| \leq |a_n| + |a_m| \leq 2M$, and the resulting inequality $|a_n^2 - a_m^2| \leq 2M |a_n - a_m|$, all correctly derived.
 - -1 : factoring identity missing or incorrect.
 - -0.5 : bound $|a_n + a_m| \leq 2M$ not justified from the boundedness of $\{a_n\}$.
- (iv) **(1 mark) Assembly.** The correct choice $\varepsilon' := \varepsilon/(2M)$ (or equivalent), applied to the Cauchy hypothesis, and the final chain of inequalities correctly matched back to the definition of $\{a_n^2\}$ being Cauchy.
 - -0.5 : N chosen depending on n (or m), rather than depending only on ε .
- (v) **(0.5 marks) Writing.** Full sentences, connectives, and inference words throughout; quantifiers written in words (“let $\varepsilon > 0$ be given,” “there exists $N \in \mathbb{N}$ such that”), never as a bare symbolic string.

Typical further deductions: -1 : uses the Cauchy Criterion or the Algebraic Limit Theorem, both explicitly banned by the question; -0.5 : sentence fragments / unjustified jumps between lines.

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